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ACS Appl. Mater. Interfaces, **Just Accepted Manuscript** • DOI: 10.1021/acsami.8b00842 • Publication Date (Web): 08 Mar 2018

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Correlating transport and structural properties in $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ (LAGP) prepared from aqueous solution

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Keywords: NASICON, synthesis, superionic conductors, structural analysis, transport properties

Abstract

$\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ (LAGP) is a solid lithium ion conductor belonging to the NASICON family, representing the solid solution of $\text{LiGe}_2(\text{PO}_4)_3$ and AlPO_4 . The typical syntheses of LAGP either involve high temperature melt-quenching, which is complicated and expensive, or a sol-gel process requiring costly organic germanium precursors. In this work, we report a simple method based on aqueous solutions without the need of ethoxide precursors. Using synchrotron and neutron diffraction, the crystal structure, the occupancies for Al and Ge, as well as the distribution of lithium were determined. Substitution of germanium by aluminum allows for an increased Li^+ incorporation in the material and the actual Li^+ content in the sample increases with the nominal Li^+ content and a solubility limit is observed for higher aluminum content. By means of impedance spectroscopy, an increase of the ionic conductivity with increasing lithium content is observed. While the lithium ionic conductivity improves, due to the increasing carrier density, the bulk activation energy increases. This correlation suggests that changes in the transport mechanism and correlated motion may be at play in the $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ solid solution.

1. Introduction

Energy storage systems are very important for many applications such as portable electronic devices, resilient storage for renewable energy sources and electric vehicles that will be of even greater importance in the future.^{1,2} As currently available battery systems are unable to fulfill the needs for higher energy and power density, prolonged cycle life and increased safety, solid electrolytes are sought after for the alternative all-solid state batteries (ASSBs).^{3,4}

Among the numerous types of solid lithium ion conductors – LISICONs, NASICONs, garnets, (anti-)perovskites and sulfide-based systems^{5–15} – NASICON materials are widely investigated.^{16–23} NASICON (Na Super Ionic CONductor) describes a family of ionic conductors with the structure of $\text{NaZr}_2(\text{PO}_4)_3$, which crystallizes in the $R\bar{3}c$ space group (*International Tables* #167).^{24,25} Apart from different sodium-containing compounds,^{26–30} there exist also stable lithium ion conductors within this structure type.^{31–33} The general formula of lithium-conducting NASICON materials is based on $\text{LiM}(\text{PO}_4)_3$ with M being a tetravalent metal ion. The basic structure consists of LiO_6 units in trigonal antiprismatic coordination, MO_6 octahedra and PO_4 tetrahedra. Li^+ occupies the Wyckoff site $6b$, referred to as Li(1) in the following, and M^{4+} the Wyckoff position $12c$, as shown in Figure 1a.³⁴ The tetravalent metal ions can be substituted by various trivalent ions such as Al, Ga, Cr, Sc, Y, Fe and La,^{31,32,35–37} which leads to the simultaneous incorporation of an additional lithium ion per formula unit for charge compensation. Thus, the ionic conductivity is increased due to higher charge carrier concentration.³⁸ The additional lithium is suggested to be located on an $18e$ position,^{27,39–41} hereafter denoted as Li(2), or split along the direction of the conduction channels around this $18e$ position on a $36f$ Wyckoff site,^{34,42–44} named Li(3) (Figure 1a). During ionic transport, Li^+ moves from the Li(1) position to the Li(2) position and from here to the next Li(1) position – or across the two Li(3) sites from one Li(1) to the other – as depicted in Figure 1b. For the jump, it has to cross two triangle areas, T1 and T2, spanned between three oxygen atoms, each.^{20,21,27,45,46} It was suggested that these triangles act as a bottleneck for ion transport, where the activation energy decreases with increasing area of the smaller triangle.^{27,47}

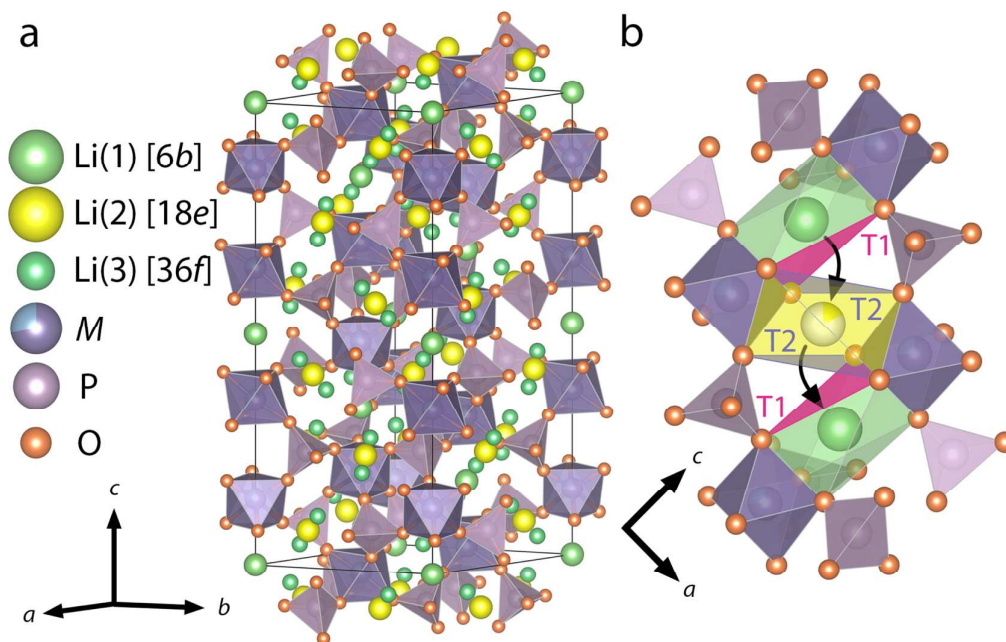


Figure 1: (a) Polyhedral representation of the crystal structure of $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ ($M = \text{Al/Ge}$). The Wyckoff 6b position for Li(1) is shown in green and the 18e position for Li(2) is depicted in yellow. The split pair site around the Li(2) position (a 36f position) is shown as smaller green spheres and referred to as Li(3). (b) Visualization of the hopping path of Li^+ through the two triangle areas T1 and T2 according to Tietz and coworkers.²⁷

$\text{Li}_{1+x}\text{Al}_x\text{Ti}_{2-x}(\text{PO}_4)_3$ and $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$, abbreviated as LATP and LAGP, respectively, are two of the most widely studied Li-conducting NASICON materials.^{48–52} These materials are relatively stable against water and air^{33,53,54} and offer bulk conductivities of up to 3 mS cm^{-1} at room temperature.⁵⁵ The ionic radii of Ge^{4+} and Al^{3+} in octahedral coordination are nearly identical, with that of Ti^{4+} being higher.⁵⁶ Thus, in order to analyze the changes in the structure without convoluting the results with a size effect of the M ion, LAGP is the more suitable material. Typically, LAGP is prepared in a costly and complicated melt-quenching process involving temperatures of over 1300°C .^{23,34,52,57–59} Another method is the Pechini-type⁶⁰ sol-gel process using germanium(IV) ethoxide, $\text{Ge}(\text{OC}_2\text{H}_5)_4$, as the germanium source.^{20,21,61–64} This precursor, however, is significantly more expensive than GeO_2 , commonly used in the melt-quenching process. For a systematic aliovalent substitution study with a multitude of samples, a facile and reasonably priced method is therefore favorable.

In this contribution, a new procedure employing the more affordable precursor GeO_2 at temperatures below 900°C out of an aqueous solution, is developed. In addition, we analyze

the structure of the resulting samples using synchrotron and neutron diffraction, which reveals a solubility limit for aluminum at $x = 0.5$ that is closely related to structural changes. Electrochemical impedance spectroscopy shows an increase of the ionic conductivity with higher Li^+ content while the activation energy turns out to increase with increasing bottleneck size, contrary to previous reports. Instead, the activation energy increases with increasing lithium content, suggesting a more complicated correlated diffusion behavior for the ion transport. This work shows that less costly synthesis routes for LAGP can be developed and that the mechanism of Li ion transport is still not fully understood and needs to be further theoretically investigated.

2. Experimental Section

Synthesis. LiOH (98 %, Acros), $\text{Al}(\text{OH})_3$ (reagent grade, Sigma-Aldrich), GeO_2 (99.999 %, ChemPur) and $\text{NH}_4\text{H}_2\text{PO}_4$ (99.9 %, Acros) were used. For precise taring, $\text{Al}(\text{OH})_3$ and $\text{NH}_4\text{H}_2\text{PO}_4$ were dried at 120 °C under vacuum to remove any residual water. LiOH was dissolved in de-ionized water. The exact Li^+ concentration was determined by titration with 0.1 M HCl (Grüssing). For the syntheses of $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$, stoichiometric amounts of the LiOH solution, $\text{Al}(\text{OH})_3$, GeO_2 , and $\text{NH}_4\text{H}_2\text{PO}_4$ were dissolved in de-ionized water, including a 5 mol% excess of LiOH to compensate for evaporation. Then, ammonium hydroxide solution (ACS reagent, Sigma-Aldrich) was added to retain a basic pH above 12. The solution was heated under constant stirring in a sand bath to evaporate the water. After that, the mixture was dried subsequently at 200 °C and 300 °C for 6 h, each. The resulting powder was ground in an agate mortar and uniaxially pressed (226 MPa) into pellets of 13 mm diameter. The pellets were placed on platinum foil in corundum crucibles, covered with a small amount of sacrificial powder and sintered at different temperatures (750 °C, 800 °C and 850 °C) with a heating rate of 200 °C/h for 8 h.

X-ray powder diffraction. X-ray diffraction measurements of the $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ pellets were carried out on a PANalytical Empyrean powder diffractometer in Bragg-Brentano θ - θ geometry with Cu K α radiation ($\lambda_1 = 1.5405980 \text{ \AA}$, $\lambda_2 = 1.5444260 \text{ \AA}$) and a PIXcel detector. The X-ray tube was operated at 40 kV and 40 mA. Measurements were carried out in the 2θ -range between 10° and 90° with a step size of 0.013°. Counting time per step was 30 s. The FullProf Suite (version June 2015) program package was utilized for Rietveld refinement⁶⁵ of the laboratory X-ray diffraction data. A pseudo-Voigt function was

used to describe the peak profiles and a linear interpolation between manually set points with refinable heights was used for the background. The atomic displacement parameters B_{iso} of Li atoms were fixed at $2.5 \text{ \AA}^2$. The second Li atom (18e) was placed at $x = 0.55$, i.e. at the position $(0.55, 0, \frac{1}{4})$, which corresponds to the center of the respective polyhedron and was kept fixed. For the refinement against the laboratory X-ray diffraction data, the occupancies of Li, Al and Ge were constrained to each other in order to preserve charge neutrality. Analysis of the refined crystal structures was performed using VESTA.⁶⁶

Neutron powder diffraction. Approximately 3 cm^3 of the $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ powder was filled into vanadium vials (diameter 1 cm) under argon atmosphere. The vials were sealed using indium wire. Neutron diffraction data were recorded in Debye-Scherrer geometry in the 2θ -range between 1° and 152° with a step size of 0.05° on the high-resolution diffractometer SPODI at Heinz Maier-Leibniz Zentrum (Garching near Munich, Germany).⁶⁷ A wavelength of $1.548171 \text{ \AA}$ was achieved by using a Ge(551) monochromator crystal with 155° take-off angle. For detection, $80 \text{ }^3\text{He}$ detector tubes (vertically sensitive, 300 mm height) were used. During refinement in the 2θ -range between 10° and 152° with TOPAS-Academic V6,⁶⁸ the peak profiles were described by a Thompson-Cox-Hastings pseudo-Voigt function.⁶⁹ A 16-coefficient polynomial was used for the background. The instrumental resolution function was obtained from a measurement of $\text{Na}_2\text{Ca}_3\text{Al}_2\text{F}_{14}$. Afterwards, the Thompson-Cox-Hastings peak shape parameters for the LAGP refinements were fixed to the values received from the standard. Size and strain formulations were used to account for sample peak broadening. At first, the structure model was refined with only the Li(1) (6b) Wyckoff site occupied. In the difference Fourier map, a peak with negative nuclear density was detected on a 36f site at about $(0, \frac{1}{4}, 0.05)$. This position is quite similar to the one found for LATP using neutron diffraction.^{42,43} Hereafter, it was used as a starting point for refinements with both lithium sites 6b and 36f occupied. The (fractional) site occupation factors N of both, the 6b and 36f, Li positions were thereby constrained to each other and to those of Al and Ge following the equations $1 = N_{\text{Li}(1)} + 6 \cdot N_{\text{Li}(2)} - 2 \cdot N_{\text{Al}}$ and $N_{\text{Al}} = 1 - N_{\text{Ge}}$, respectively, in order to retain overall charge compensation. Atomic displacement parameters of both Li sites were refined isotropically and kept identical, while those of the other atoms were refined in anisotropic approximation using a second order tensor.

Synchrotron powder diffraction. High-resolution synchrotron powder diffraction data were collected in Debye-Scherrer geometry at beamline 11-BM at the Advanced Photon Source, Argonne National Laboratory, Lemont, IL, USA. The samples were filled into Kapton

capillaries (inner diameter of 0.80 mm) and sealed using modeling clay. Diffraction patterns were obtained in the 2θ -range between 0.5° and 50° (step size 0.001°) with a calibrated wavelength of $0.414534 \text{ \AA}$ (Si(111) double crystal monochromator). For detection, 12 independent Si(111) analyzer crystals spaced 2° from each other were used in combination with LaCl_3 scintillation detectors. Rietveld refinements were carried out in the 2θ -range between 2° and 40° , using FullProf Suite⁶⁵ (version June 2015). Instrumental resolution parameters were obtained from an LaB_6 standard. The peak profiles were described by the convolution of a Thompson-Cox-Hastings pseudo-Voigt function⁶⁹ and a function for axial divergence asymmetry.⁷⁰ For the background, a linear interpolation between manually set points with refinable heights was used. Anisotropic strain broadening was described by a quartic function.⁷¹ Due to the low X-ray scattering form factor of lithium, positions and atomic displacement parameters for the lithium atoms were not refined. While the atomic displacement parameters B_{iso} of the Li atoms were fixed to $1.5 \text{ \AA}^2$, those for the other atoms were refined anisotropically. The second lithium atom was placed at $(0, \frac{1}{4}, 0.05)$, the Li(3) site obtained from the neutron refinements. The occupancy for Li(3) was fixed to the values obtained from neutron refinements of samples with the same nominal composition.

For the refinement against synchrotron X-ray data, the occupancies of Li, Al and Ge were constrained to each other in order to preserve charge neutrality. Analysis of the refined crystal structures was performed using VESTA.⁶⁶

Bond valence sum analysis. In order to predict the preferential lithium diffusion pathways, the bond valence sum method (BVS) was used. It has already been employed successfully for various lithium ion conductors.^{43,72} This method assumes that ion diffusion within a solid from one Li site to an adjacent one occurs along a path, on which the valence mismatch ΔV calculated by

$$\Delta V = |V - V_{\text{nom}}| \quad (1)$$

between the bond valence V and the nominal valence V_{nom} of the ion is minimal.⁷³

Hereby, the bond valence sum for a cation A is given as

$$V = \sum_X S_{\text{A-X}} \quad (2)$$

with $S_{\text{A-X}}$ being the bond valence between A and the anions X, which in turn can be calculated from the bond lengths $R_{\text{A-X}}$ according to

$$S_{\text{A-X}} = \exp\left(\frac{R_0 - R_{\text{A-X}}}{b}\right) \quad (3)$$

using the empirical parameters R_0 and b .⁷³ Calculations were performed using tabulated

values for R_0 and b as well as the positions of Ge, P and O^{43,73}.

Electrical conductivity measurements. Gold electrodes (99.99 %, ChemPur) with a diameter of 8 mm were vapor deposited (Sputter Coater, tectra GmbH) on both sides of the $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ pellets. Aluminum current collectors were pasted to the contacts. Electrochemical impedance spectroscopy was performed using an Alpha-A high performance frequency analyzer with a ZG4 impedance test interface (Novocontrol) operating in 2-point mode. The AC amplitude was 10 mV, 20 MHz to 100 mHz was chosen as frequency range. Measurements were performed between 0 °C and 50 °C (± 0.3 °C) in steps of 2.5 °C, using a Novocool cryostat (Novocontrol) via nitrogen gas flow and electrical heating elements. Analysis of the obtained data was done using RelaxIS 3 (rhd instruments).

3. Results and Discussion

Synthetic optimization for high conductivity. As an initial step, the sintering temperature for the new synthesis method was optimized for three different compositions in $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ (LAGP) with $x = 0.3, 0.5$ and 0.7 , respectively. Figure 2a shows the refined additional Li^+ in the structure and the LAGP phase fraction of the samples as function of the nominal composition for different sintering temperatures between 750 °C and 850 °C, as determined from Rietveld refinements of laboratory X-ray diffraction measurements. Two exemplary refinements are depicted in Figure S1. The overall Li^+ that is incorporated was obtained from the refined occupancy of Al^{3+} in the structure, as an increase in Al^{3+} content must lead to direct charge compensation with an increase in the Li^+ content, which can indeed be observed. However, at higher nominal Al^{3+} contents, the detectable content of Al^{3+} (and hence Li^+) incorporated in the structure decreases, suggesting some solubility limit to be reached. Beyond this point, additional lithium and aluminum are not incorporated into the structure and form side phases, leading to a decrease of the LAGP mass fraction and an increasing mass fraction of impurity phases (see Figure 2a). Comparing the different sintering temperatures, it can be observed that the samples sintered at 800 °C provide both the highest Al^{3+} content and the highest purity. At 750 °C, the phase purity is lower because of the preferred formation of $\text{Li}_9\text{Al}_3(\text{P}_2\text{O}_7)_3(\text{PO}_4)_2$, which is known to already form at lower temperatures.⁷⁴ At 850 °C, however, the evaporation of lithium is likely promoting a reduced lithium content in LAGP, as well as higher fraction of side phases due to the resulting lack of stoichiometry. Previous studies also showed a higher fraction of impurity phases for sintering temperatures above 800 °C in the case of LAGP glass-ceramics.^{54,58}

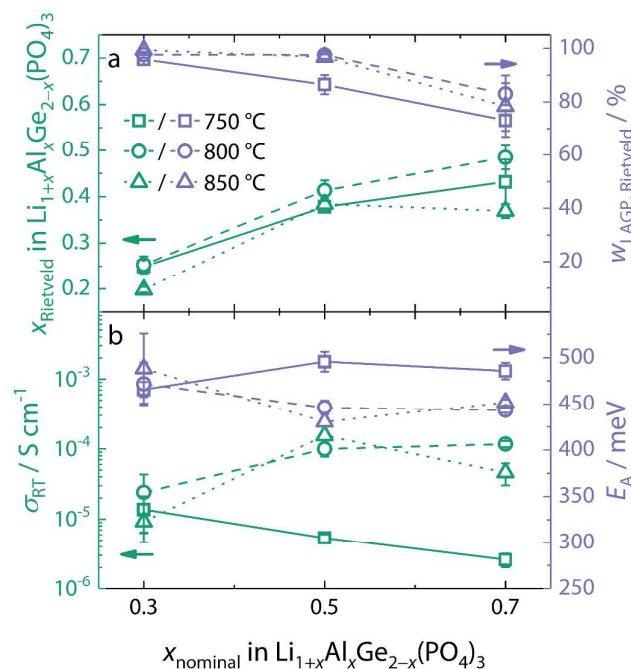


Figure 2: (a) Variation of the additional $x(\text{Li}^+)$ in $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ (LAGP) as determined by Rietveld refinements against laboratory X-ray diffraction data (x_{Rietveld} , green) and mass fraction of LAGP in the samples ($w_{\text{LAGP, Rietveld}}$, violet) for different nominal Li^+ contents x_{nominal} . (b) Room temperature total conductivity σ_{RT} (green) and total activation energy E_A (violet) for different nominal Li^+ contents. Data points depict the mean value of different samples with identical nominal composition; the shown uncertainty reflects the standard deviation. Lines are given as a guide to the eyes. The samples sintered at 800 °C provide the best combination of high purity and high Li^+ concentration, which results in the highest conductivity and lowest observed activation energy.

In addition to the phase and structural analysis, Figure 2b shows the ionic conductivity and total activation energy obtained from impedance spectroscopy of the LAGP samples that were prepared at different temperatures. Because of the highest lithium content in the structure and highest phase purity, the samples sintered at 800 °C exhibit the highest conductivities and lowest activation energies. The samples sintered at 850 °C are very similar because of a slightly higher density, which seems to compensate for the lower lithium content. For the samples sintered at the lower temperature of 750 °C the density is severely decreased (not shown) and, thus, the total activation energy is higher and the resulting conductivity lower. All data points shown in Figure 2 represent the mean values of multiple samples measured per data point and the uncertainty values given are their standard

deviations, showing the high reproducibility of the newly developed synthesis method.

In conclusion, a sintering/reaction temperature of 800 °C turns out to be optimal for high phase purity and best ionic conductivity. For a more detailed analysis of the crystal structure and transport in $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$, samples with a smaller variation of the Al^{3+} content in the nominal compositions were sintered at 800 °C consequently, and are analyzed in the following.

Composition-dependent structural evolution. $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ was synthesized with $x = 0.0, 0.2, \dots 0.8$, and a sintering temperature of 800 °C to obtain samples with the highest LAGP phase fraction possible. Synchrotron and neutron diffraction data were collected for all nominal compositions. Representative diffraction data and refinements for samples with $x_{\text{nominal}} = 0.4$ are shown in Figure 3. All results shown are obtained from synchrotron and neutron diffraction measurements of one sample of each composition.

From the Rietveld refinements against synchrotron and neutron diffraction data, x_{Rietveld} as a measure for the apparent lithium composition and the mass fraction of LAGP ($w_{\text{LAGP, Rietveld}}$) are obtained. The evolution of these parameters with changing nominal lithium composition is shown in Figure 4a. At first, x_{Rietveld} increases almost linearly with a slope of unity with increasing nominal composition (x_{nominal}), showing that all added aluminum is indeed incorporated into the structure. In this initial x -range, the phase fractions remain constant with a mass fraction of LAGP close to 100 %. However, starting at $x_{\text{nominal}} \approx 0.5$, the determined Al^{3+} occupancy begins to saturate and cannot be increased any further. This saturation of Al^{3+} in the structure suggests a solubility limit of $x_{\text{nominal}} \approx 0.5$ and higher Al^{3+} contents lead to a strong decrease in the LAGP phase fraction. The observed side phases include GeO_2 ($P3_121$), AlPO_4 ($C222_1$), $\text{Li}_4\text{P}_2\text{O}_7$ ($P\bar{1}$) and $\text{Li}_9\text{Al}_3(\text{P}_2\text{O}_7)_3(\text{PO}_4)_2$ ($P\bar{3}c1$).⁷⁵ The latter is preferably formed at higher aluminum contents, as also shown by Francisco *et al.*²¹

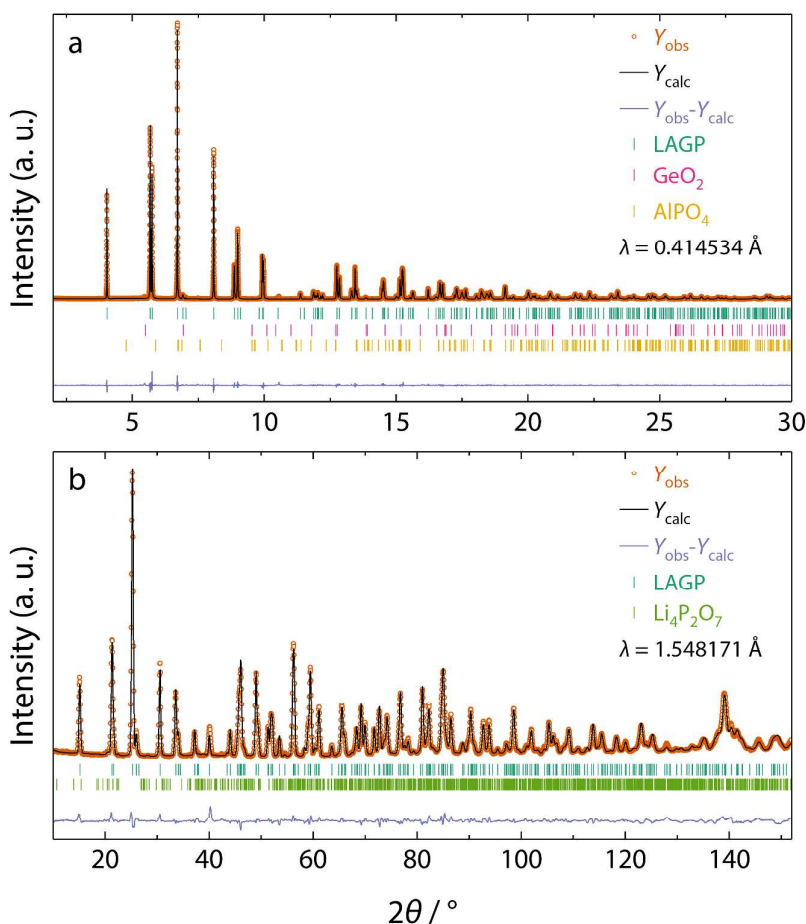


Figure 3: Representative synchrotron (a) and neutron (b) diffraction data (open circles) for two LAGP samples with nominal composition $\text{Li}_{1.4}\text{Al}_{0.4}\text{Ge}_{1.6}(\text{PO}_4)_3$ ($x = 0.4$), sintered at 800 °C. The calculated pattern obtained from the refinement is shown as a black line, the violet line depicts the profile difference. The calculated Bragg positions of the refined phases are marked with vertical ticks: (a) Refined mass fractions: 99.6(2) % $\text{Li}_{1.43}\text{Al}_{0.43}\text{Ge}_{1.57}(\text{PO}_4)_3$, 0.31(1) % GeO_2 , 0.11(1) % AlPO_4 ; goodness of fit $S = 1.73$, $R_p = 4.96$ %, $R_{wp} = 6.95$ %, $R_{exp} = 4.01$ %; (b) refined mass fractions: 98.5(3) % $\text{Li}_{1.40}\text{Al}_{0.40}\text{Ge}_{1.60}(\text{PO}_4)_3$, 1.5(3) % $\text{Li}_4\text{P}_2\text{O}_7$; goodness of fit $S = 4.17$, $R_p = 4.61$ %, $R_{wp} = 5.95$ %, $R_{exp} = 1.42$ %.

Comparing the evolution of the lattice constants and cell volume (Figure 4c) to that of the lithium content, one can notice a similar behavior. As reported before,⁶¹ the lattice parameter a seems to remain constant within the resolution limits, because the ionic radii of Al^{3+} (0.535 Å) and Ge^{4+} (0.530 Å) in an octahedral coordination are nearly the same.⁵⁶ The lattice parameter c , and with it the lattice volume V , increase for lithium contents up to $x = 0.5$. For even higher nominal lithium contents, c and V exhibit the same saturation behavior as the lithium content, which suggests that the added lithium is directly responsible for the unit cell expansion. These results are comparable to earlier work on LAGP,⁶¹ in which a solubility

limit for $x = 0.6$ was found.

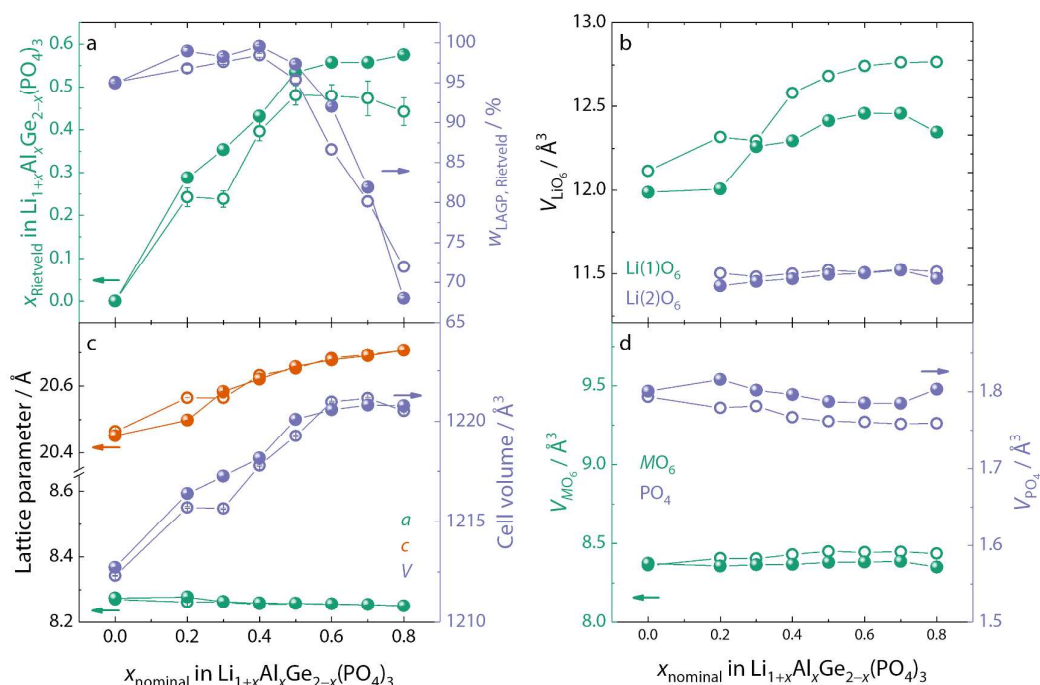


Figure 4: Dependence of $\text{Li}^+/\text{Al}^{3+}$ composition (x_{Rietveld}) and mass fraction of $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ ($w_{\text{LAGP, Rietveld}}$) (a), lattice constants a , c and cell volume V (c), volume of LiO_6 octahedra (b) as well as volume of MO_6 ($M = \text{Al, Ge}$) octahedra and PO_4 tetrahedra (d) on nominal lithium content (x_{nominal}). All values were determined by Rietveld refinement of synchrotron X-ray diffraction data (closed circles) or neutron diffraction data (open circles). The observable Li^+ content increases with the nominal value of x , but reaches a solubility limit at $x_{\text{Rietveld}} \approx 0.5$. The increase in unit cell volume correlates with the increase of x_{Rietveld} and reaches saturation at the same nominal lithium content. This increase can also be seen for the volume of the Li(1)O_6 octahedra (b), whereas the volume of the other polyhedra remains mostly constant (d).

As an explanation for the expanding unit cell, Figure 4b and 4d show the dependence of the LiO_6 and MO_6 octahedra volumes as well as the PO_4 tetrahedra volumes as a function of the nominal lithium content. The largest change is visible for the Li(1)O_6 octahedron, *i.e.* the Wyckoff 6b site. All other polyhedral volumes remain relatively constant. The increase of the c -axis and the unit cell volume directly seem to correlate with the increase of the Li(1)O_6 octahedral volumes, as these directly point in the z -direction (see Figure 1). Substitution of germanium by aluminum does not lead to a significant change in the MO_6 octahedra volume. Interestingly, the addition of Li, which has to be placed on the 18e or 36f position (the 6a site is already fully occupied for $x = 0$) does not affect the Li(2)O_6 volume much, but only

increases the Li(1)O₆ volume. This behavior may possibly be explained by the reduced charge on the adjacent Wyckoff 12c site due to the substitution of Ge⁴⁺ by Al³⁺, in combination with the additional charge localized on or around the 18e Li(2) site. The Li(1) site shares a face with the (Ge/Al)O₆ octahedra along the *z*-direction and now lithium can move further in *c*-direction as a result of weaker Coulombic repulsion in this direction. While this alone will not result in larger volume, an increasing occupancy of the adjacent Li(2) or Li(3) positions, resulting in an increase of the Coulombic repulsion might explain the increasing Li(1)O₆ polyhedra volumes, as this behavior will minimize the Coulombic repulsion.

In order to further analyze the Li⁺ distribution on the different Li sites (6b, 18e and 36f), the collected neutron diffraction data are helpful. In the difference Fourier maps of refinements with only the Li(1) site occupied (see Figure S2), high negative nuclear density (corresponding to scattering lithium) was found on a 36f Wyckoff site at about (0, ¼, 0.05). This Li(3) position is close to the 18e Li(2) site, but splits along the Li⁺ conduction pathway between an Li(1) and an Li(2) site. For LATP, a similar 36f site was previously determined using neutron diffraction.^{42,43} In the case of *x* = 0 (the sample without any aluminum), Li was only found on the 6b site. Thus, for further analyses with *x* > 0, Li was refined on the Li(1) and Li(3) positions. The occupancies of both lithium sites were refined, while their sum was constrained to those of Al and Ge on the *M* site for charge compensation. The obtained lithium occupancies are shown in Figure 5a. It can be seen that, as soon as any additional lithium is added to LiGe₂(PO₄)₃, redistribution from the Li(1) to the Li(3) site occurs. This redistribution was already reported for sodium containing NASICONs⁷⁶ as well as LATP.⁴² In addition, Francisco *et al.* observed similar behavior in LAGP using synchrotron X-ray data.²¹ In this work, we arrive at the same conclusion using neutron diffraction data, which is better suited for the analysis of the Li occupancy in materials. The redistribution of lithium from the Li(1) to the Li(3) site is said to reduce the newly created Li–Li repulsions due to the inclusion of additional Li, which cannot be placed on an Li(1) site anymore.^{21,42} However, with a reduction of the repulsion, the increasing polyhedral volume remains puzzling.

In addition to the occupation analysis, bond valence mismatch calculations were performed in order to visualize the Li⁺ conduction pathways. According to this method, conduction in solids happens along channels on which the difference between bond valence and nominal valence is minimized.^{43,77} The resulting bond valence isosurface for a sample with nominal composition Li_{1.5}Al_{0.5}Ge_{1.5}(PO₄)₃ is shown in Figure 5b. Conduction occurs along zigzag channels, which contain the Li(1) and Li(2) sites, similar to the situation in

LATP.⁴³ The Li(3) site is located at the bends of the channels, which confirms the position found using difference Fourier maps. Figure 5c shows the conduction pathway in the bottleneck areas. Obviously, lithium ions have to cross the two triangle-shaped areas in order to be transported through the material.

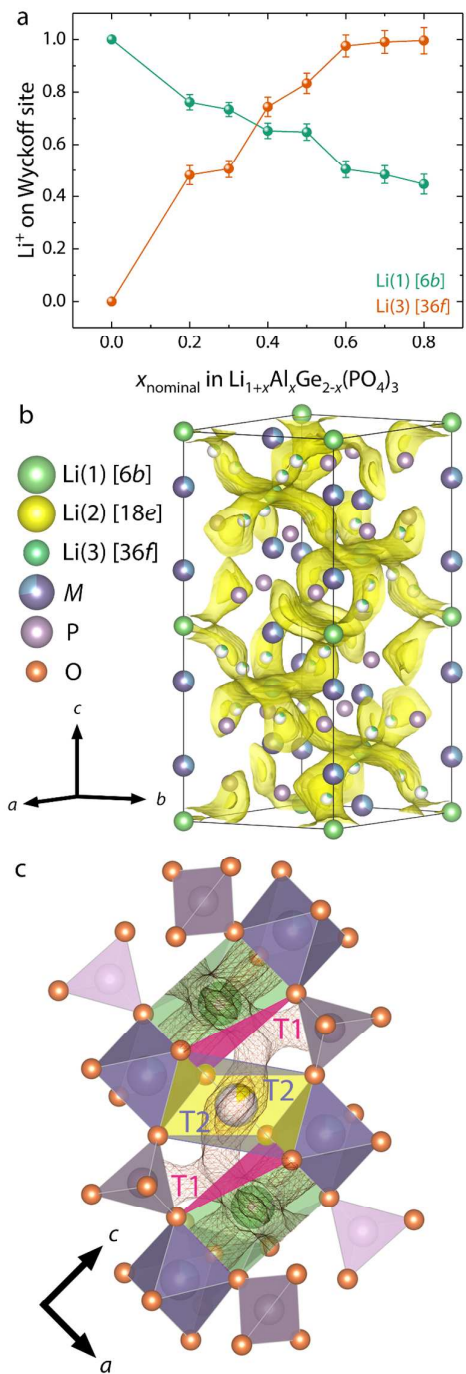


Figure 5: (a) Distribution of Li⁺ ions on the Li(1) (6b) and Li(3) (36f) positions in the $\text{Li}(1)_y\text{Li}(3)_z\text{Al}_{y+z-1}\text{Ge}_{3-y-z}(\text{PO}_4)_3$ structure, where y is the fraction on the 6b site and z that on

the 36f site, as a function of nominal additional lithium (x_{nominal}). The occupancies were determined by Rietveld refinement against neutron diffraction data. It is evident that the addition of lithium leads to redistribution from the Li(1) to the Li(3) site. (b) Bond valence mismatch illustrating potential conduction pathways across a unit cell in $\text{Li}_{1.5}\text{Al}_{0.5}\text{Ge}_{1.5}(\text{PO}_4)_3$. (c) Diffusion pathway (shown as a red grid for clarity) across the triangle-shaped bottleneck areas determined from the bond valence mismatch.

Evolution of transport properties. Pellets of the materials with different compositions were analyzed by impedance spectroscopy in order to examine the ionic transport properties. Figure 6 shows representative impedance data for a sample with nominal composition $\text{Li}_{1.6}\text{Al}_{0.6}\text{Ge}_{1.4}(\text{PO}_4)_3$. Apart from the obtained conductivity values, the impedance responses are similar along the series of solid solutions. A series of two RQ elements (a resistance R and a constant phase element Q in parallel) and a constant phase element are used to fit the obtained spectra consisting of two semicircles and a tail at low frequencies. The high-frequency semicircle with capacitances of about $2 \cdot 10^{-11} \text{ F cm}^{-2}$ is attributed to intra-grain conduction, *i.e.* bulk conductivity, the other semicircle at lower frequencies ($C/A \approx (0.2 - 1) \cdot 10^{-9} \text{ F cm}^{-2}$) to conduction through the grain boundaries.⁷⁸ Because the gold electrodes are unable to deliver lithium a polarization feature is visible at low frequencies. Data of the temperature-dependent impedance measurements are used to create the Arrhenius plots in Figure 6a. Here, the reciprocal resistance for the bulk and grain boundary contribution as well as the reciprocal of the sum of those resistances was used instead of the conductivity. This is due to the fact that the thickness of the grain boundaries, which is necessary to calculate the grain boundary conductivity,⁵⁴ is unknown. Figure 6b shows a Bode-plot of the impedance data to support the employed equivalent circuit and show the quality of the fit.

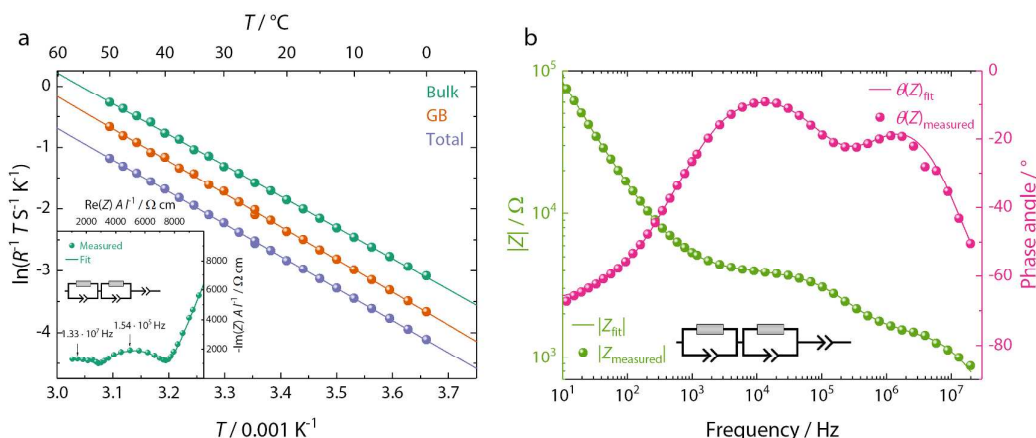


Figure 6: (a) Arrhenius plot for a sample with nominal composition $\text{Li}_{1.6}\text{Al}_{0.6}\text{Ge}_{1.4}(\text{PO}_4)_3$

sintered at 800 °C, as obtained from temperature-dependent impedance spectroscopy. A Nyquist plot of the data measured at 25 °C (circles), together with the fitted data (line) according to the displayed equivalent circuit, is shown as an inset. The high-frequency semicircle was assigned to bulk resistance, the semicircle at intermediate frequencies to grain-boundary resistance. A polarization feature at low frequencies was modelled by using a constant-phase element due to the gold electrodes. The Arrhenius plot shows the temperature dependence of the different processes. (b) Bode plot showing the measured data of the same sample at 25 °C and the high quality of the corresponding fit.

Figure 7 shows the obtained ionic conductivity and activation barrier of the total and bulk ionic processes along the series of solid solutions. The conductivity increases with increasing charge carrier density, *i.e.* Li⁺ ion content until $x \approx 0.5$ leading to a similar trend as reported by Francisco *et al.*²¹ As seen above, at higher Al³⁺ contents a solubility limit is reached and no additional Li⁺ or Al³⁺ are incorporated into the structure. While the stagnant concentration of charge carriers should lead to a plateau of the conductivity, a decrease is observed for higher Al³⁺ contents. This decrease in conductivity stems from the decreased phase fraction of LAGP and the formation of non-conducting side phases, as shown in Figure 4a. However, the activation energy of the total conductivity decreases for increasing lithium content due to an increase in density (see Supporting Information Figure S3), which leads to a decreased fraction of grain boundaries that have the higher activation energy in comparison to the bulk contribution. At $x > 0.5$, the activation energy of the total conductivity remains constant, possibly because of a compensation effect of increasing density and decreasing LAGP phase fraction.

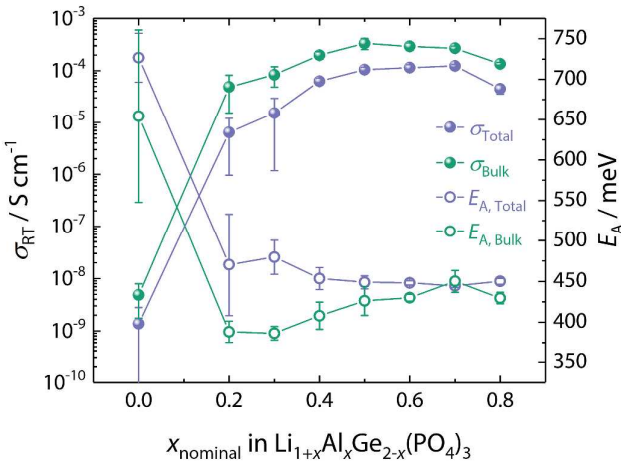


Figure 7: Room-temperature conductivity σ_{RT} (filled circles) and activation energy E_{A} (open circles) as a function of nominal Li⁺/Al³⁺ content (x_{nominal}) in $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$. Values for

bulk conduction are shown in green, the total conductivity and total activation energy is displayed in violet. All data points are calculated as average values from multiple samples with identical nominal composition, and the uncertainties reflect the standard deviation. Lines are given as a guide to the eyes. The conductivity increases with increasing Li^+ content until a saturation is reached at $x_{\text{nominal}} \approx 0.5$. This correlates with the saturation of the actual Li^+ content (see Figure 4a). At even higher nominal contents, the conductivity decreases again because of a severe decrease in LAGP phase fraction (see Figure 4a). The total activation energy decreases with increasing x , because of increasing density and thus reduced grain boundary resistance. However, the bulk activation barrier increases at higher x .

In contrast to the total conductivity that includes grain boundary effects, the bulk activation energy exhibits a different behavior. At first, it decreases, but only until $x = 0.3$. With further increasing lithium content, the $E_{\text{A,Bulk}}$ increases again. A similar trend was reported by Francisco *et al.* with the minimum shifted to lower x .²¹ These data show that it is important to distinguish between bulk and grain boundary transport as both can follow completely different trends.

Relationship between structure and transport. In order to understand the transport properties reported above, the occurring structural changes need to be considered. Figure 8 shows that the conductivity increases with the volume of both LiO_6 octahedra. This may be because of the increasing width of the diffusion pathways for the lithium ions, as well as the increasing nominal carrier concentration that leads to the increasing polyhedral volumes in the first place. The activation energy, however, also increases with increasing octahedral volumes. Calculations for different compounds derived from $\text{LiTi}_2(\text{PO}_4)_3$ predicted an opposing trend: a decrease of activation energy with increasing polyhedron volume is usually expected.⁴⁰ Furthermore, in contrast to previous reports for sodium-based NASICONs²⁷ as well as LAGP,²¹ the activation energy also increases with the area of the triangles the Li^+ ions are moving through (Figure 8c). However, the here-observed trend of bulk activation energy versus nominal lithium concentration closely resembles the one reported by Francisco *et al.* for LAGP.²¹ In the work by Francisco *et al.*, the authors suggest that the activation energy is the sum of the energy difference between the Li(1) and Li(2) site as well as the actual jump barrier height that has to be overcome, with the energy difference decreasing exponentially with higher lithium content due to redistribution between the two lithium sites. The increasing barrier height was attributed to an increase in bottleneck area, which itself was

obtained from peak broadening via Raman spectroscopy. Unfortunately, it was not possible to differentiate between the influence of bottleneck size and lithium distribution on the peak width, which suggested a mixture of both effects.²¹ In this work, the bottleneck area is determined by synchrotron as well as neutron diffraction data, both of which are methods that are able to determine atomic positions, and thus the triangle area, accurately. The here-presented data suggest that the increase of the bottleneck area is not the determining factor for the activation energy in $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$. Instead, a higher activation energy is measured for samples with larger bottleneck area and changes in the diffusion mechanism may have to be considered as possible explanation.

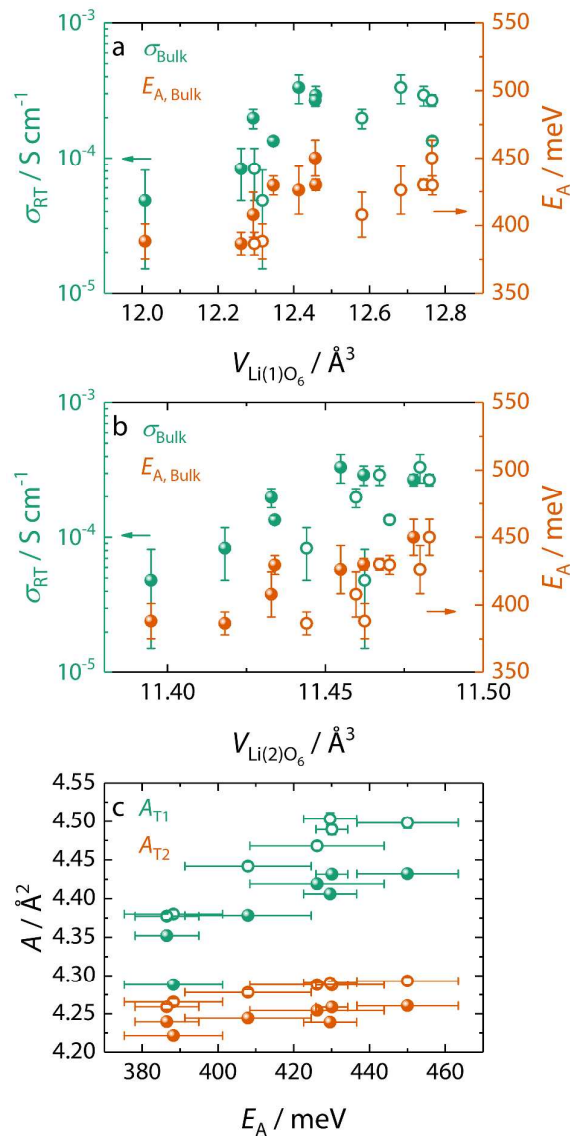


Figure 8: Dependence of room-temperature conductivity σ_{RT} and activation energy E_A on the volume of the Li(1) (a) and Li(2) (b) octahedra (determined from Rietveld refinements of synchrotron X-ray (filled circles) and neutron (open circles) diffraction data) as well as

dependence between the triangle areas of both Li polyhedra and the bulk activation energy (c). Conductivities and activation energies depict the mean values of multiple samples with identical nominal composition; error bars reflect the standard deviation. Errors of triangle areas resulted from the refinements. The conductivity and bulk activation energy increases with increasing octahedral volume, likely due to the increasing Li^+ concentration. In contrast to the typical understanding, the activation energy increases with an increasing area of the bottleneck triangle.

Indeed, the diffusion mechanism in the material may need to be questioned. According to the classical diffusion model, individual ions jump uncorrelated from one site to another. In that assumption, only the lattice arrangement, *i.e.* the structure, determines the energy landscape for this process.^{79,80} However, a recent publication by He *et al.* has shown that this theory is unable to explain the extremely low activation energies found in certain superionic conductors.⁸⁰ In the case of superionic conduction, a concerted ion migration mechanism has to be considered. Using molecular dynamics simulations, the authors predicted that two neighboring lithium ions migrate as a pair in $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$. During that process, a lithium ion from the high-energy Li(2) site moves to a Li(1) site at lower energy. This correlated motional process partly eliminates the energy barrier for the uphill migrating ion, resulting in a lower activation energy than that dictated by the crystal structure. In particular, a Li sublattice ordering enables this process,⁸⁰ thus explaining the Li redistribution shown in Figure 5a. A similar concerted migration process was also reported for $\text{LiZr}_2(\text{PO}_4)_3$, lately.⁸¹ If the aforementioned model can be transferred to our experimental data for LAGP, it suggests that for $x = 0.3$ the optimum sublattice ordering might be achieved, because this is the composition with the lowest measured bulk activation energy. For higher lithium concentrations, the Li(1) site occupancy is lower and that of the Li(3) site higher, and thus the migration of the Li(1) ion to the Li(3) site may be more hindered. Additionally, He *et al.* further reported a flat energy landscape around the high-energy Li(2) site, leading to an extended occupancy density for lithium in this region.⁸⁰ This correlates with the here-observed occupied Li(3) site as determined from the neutron diffraction data, which represents a split site around the Li(2) position. These structural and transport data show that correlated motion may be prevalent not only in $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ ⁸⁰ and $\text{LiZr}_2(\text{PO}_4)_3$,⁸¹ but also in LAGP. In order to corroborate our findings, extensive ab-initio molecular dynamics simulations need to be performed.

4. Conclusions

In this work, a novel synthesis for $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ has been explored and the changing structure due to Al^{3+} incorporation, as well as its effect on the ionic transport has been examined. Neutron and synchrotron diffraction data show an increase in the unit cell dimensions with increasing lithium content up to a solubility limit for aluminum at $x = 0.5$. Up to this point, the additional lithium was found to be incorporated on the $36f$ site that becomes increasingly populated during a concurrent depopulation of the $6b$ site.

Using impedance spectroscopy, the structural data can be linked to the material's ionic conductivity. An increase in ionic conductivity with increasing Li^+ content was observed while the total activation energy decreased. However, this work shows that it is necessary to distinguish between bulk and grain boundary contributions, as these exhibit opposite behavior. While the activation energy for ion transport across grain boundaries – and with it the total activation energy – decreases for higher lithium contents, the bulk activation barrier increases with increasing x , due to occupancy changes between the two lithium positions. In accordance with theoretical work for LATP,⁸⁰ this work suggests that not the structural changes and the triangle-shaped bottleneck in the diffusion pathway determine the activation barrier in LAGP, but instead correlated ion motion may be at play.

ASSOCIATED CONTENT

Supporting Information

The results of the Rietveld refinements of synchrotron and neutron diffraction data, difference Fourier maps and composition-dependent pellet density can be found here. Further, the .cif files of all compositions, obtained from Rietveld refinements against neutron diffraction data are included.

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Notes

The authors declare no competing financial interests.

ACKNOWLEDGMENT

The authors acknowledge financial support by BASF SE within the International Network for

Electrochemistry and Batteries. W.G.Z furthermore gratefully acknowledges the financial support through start-up funding provided by the Justus-Liebig-University Giessen. This work is based upon experiments performed at the SPODI instrument operated by FRM II at the Heinz Maier-Leibnitz Zentrum (MLZ), Garching, Germany. Use of the Advanced Photon Source at Argonne National Laboratory was supported by the U. S. Department of Energy, Office of Science, Office of Basic Energy Sciences, under Contract No. DE-AC02-06CH11357.

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